



Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Bachelor of Technology (Computer Science and Engineering)

Programme Code: BTECCE

Pattern 2025

With Effect from Academic Year 2025 – 2026

	Authority	Date
Proposed by	Board of Studies in Engineering Sciences	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Programme Structure

Faculty	Science and Technology	Pattern (AY-2025-2026)	2025
Department	Engineering Science	Date (w.e.f.)	04/08/2025
Programme	BTech Computer Science and Engineering		

SEM-I

Course Type / NEP Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Total Marks
			L	T	P		
BSC	BTECCE25101	Engineering Calculus and Applications	3	1	0	4	100
ESC	BTECCE25102	Fundamentals of Electronics	3	0	2	4	100
ESC	BTECCE25103	Computer Programming Principles	3	0	2	4	100
VSEC	BTECCE25104	Web Application Development	1	0	2	2	100
BSC	BTECCE25105	Applied Science and Engineering Project I	0	0	4	2	100
CC	BTECCE25106	Value Added Career Orientation I	2	0	0	2	100
IKS	IKS25CE01A	Indian Knowledge System	2	0	0	2	100
AU	BTECCE25107	Induction Training			1*	0	
Total			14	01	10	20	700

Head of the Department**Dean**

*ERP to provide document upload facility for evidencing – BoS meeting MoM, AC MoM, final approved grid with BoS chairman signature, etc.

BTECCCE25101: Engineering Calculus and Applications

Course Type	BSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03Hr./Week	Lecture	03	CIE Marks	
Tutorial	01Hr./Week	Tutorial	01	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	04Hr./Week	Total	04		

Course Description

This Applied Mathematics course will explore the practical applications of calculus, focusing on topics like applications of derivatives, successive differentiation, partial derivatives, ordinary differential equations, multiple integrals, and Fourier series. It aims to equip students with the mathematical tools to model and solve problems in various scientific and engineering fields.

Course Outcomes

CO No.	Statement
1	Apply the knowledge of differential calculus to solve problems arising in various fields of engineering and technology
2	Learn the notion of partial differentiation to calculate rates of change of multivariate functions and Applications
3	Solve ordinary differential equations; apply the knowledge of the same related to simple electrical circuits, orthogonal trajectory, and Newton's Law of cooling.
4	Evaluate double & triple integrals. Apply knowledge of multiple integrals to find area and volume.
5	Calculate the Full range and half range Fourier series, and apply Fourier series in harmonic analysis and applications.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2	2	1			1						1	1			1	1,2,3
CO2	2	1							2			1	1			1	2,3
CO3	3	1		1								1	1			1	1,2,3
CO4	3	2	1	2		1			1			1	1			1	1,3
CO5	3	2	1	2		2			1			1	1			1	1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	DIFFERENTIAL CALCULUS	08 Hours	CO	BTL
Introduction to limits and continuity, Successive differentiation, Leibnitz's Theorem for n^{th} derivatives of the product of two functions, Rolle's theorem, Lagrange's and Cauchy's mean value theorem, Taylor's and Maclaurin's series expansion.			1	1,2,3
UNIT NO.02	PARTIAL DERIVATIVES	10 Hours		
Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications-Jacobians, functionally dependence/independence.			2	2,3
UNIT NO.03	ORDINARY DIFFERENTIAL EQUATIONS	10 Hours		
Definition, Order & Degree of DE., Exact differential equations, Equations reducible to exact equations, Linear differential equations, Reducible to linear differential equations (Bernoulli's equation). Applications of ODE -Orthogonal trajectories, Newton's law of cooling, Law of Decay, Electrical Circuits etc.			3	1,2,3
UNIT NO.04	MULTIPLE INTEGRATION	07 Hours		
Introduction to 2D and 3D coordinate systems, Evaluation of double integration, Evaluation of double integrals by changing the order of integration and by changing into polar coordinates, Applications of Double integration. Introduction to triple integration. Applications			4	1,3
UNIT NO.05	FOURIER SERIES AND HARMONIC ANALYSIS	10 Hours		

FOURIER SERIES: Periodic functions, Dirichlet's condition, Fourier Series of periodic functions with period 2π and with arbitrary period $2L$, Odd and even functions, Half-range Fourier series, Harmonic Analysis, Applications.	5	1,2,3
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Textbooks

1	Higher Engineering Mathematics, B.S. Grewal, Khanna Publishers, 35th Edition, 2000.
2	Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill New Delhi, 11 th Reprint, 2010.

Reference Books / Journal Articles / Weblink

Reference Books	
1	Applied Mathematics (Vol I & II) by P. N. Wartikar & J. N. Wartikar
2	Advanced Engineering Mathematics, 7 th Edition, By Peter V. O'Neil (Thomson learning publisher)
3	Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.

Relevant Springboard Courses

Unit	Topics	Springboard Course	CO
1. Differential Calculus	Introduction to limits and continuity, Successive differentiation, Leibnitz's Theorem for n^{th} derivatives of the product of two functions, Rolle's theorem, Lagrange's Mean Value Theorem, Cauchy's Mean Value Theorem, Taylor Series Expansion, Maclaurin Series Expansion	<u>Calculus: Getting Started with Derivatives</u>	1
2. Partial Derivatives	Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Chain Rule, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications, Jacobians, functional dependence/independence	<u>ML Algorithms: Multivariate Calculation & Algorithms</u>	2

3. Ordinary Differential Equations	Definition, Order & Degree of DE, Exact differential equations, Equations reducible to exact equations (type-1), Equations reducible to exact equations (type-2), Linear differential equations, Reducible to linear differential equations (Bernoulli's equation). Applications of ODE-Orthogonal trajectories, Newton's law of cooling, Law of Decay, Electrical Circuits etc.	<u>Differential Equation</u> , Worksheet for Differential Equations	3
4. Multiple Integration	Introduction to 2D and 3D coordinate systems, Evaluation of double integration, Evaluation of double integrals by changing the order of integration, Evaluation of double integrals by changing into polar coordinates, Applications of Double integration, Introduction to triple integration, Applications	<u>Calculus: Understanding Integration</u> <u>Solving Integral Calculus</u>	4
5. Fourier Series and Harmonic Analysis	Periodic functions, Dirichlet's condition, Fourier Series of periodic functions with period 2π , Fourier Series of periodic functions with arbitrary period $2L$, Odd and even functions, Half-range Fourier series, Half-range Fourier series odd extension, Half-range Fourier series even extension, Harmonic Analysis, Applications.	<u>Understanding Fourier Analysis</u> <u>Fourier: Making Waves</u>	5

List of Tutorials

Sr. No	Name of Tutorial
1	nth Order Derivatives: Sensor Response Modeling Applications
2	Mean Value Theorem: Understanding Average System Behaviors
3	Series Expansion and Continuity: Approximating Embedded System Functions
4	Multivariable Calculus: Performance Analysis for Multicore Processors
5	Composite and Total Derivatives: Layered Control System Dependencies
6	Homogeneous Functions and Jacobians: Transformation Uses in Robotics
7	Jacobian Uses: Analyzing System Variable Dependencies
8	Exact Differential Equations: Signal Dynamics and Control Systems
9	Linear and Bernoulli Equations: Power and Energy Flow Modeling
10	Orthogonal Trajectories and Cooling Laws: Thermal Design in Microchips
11	Exponential Decay and Circuits: Transient Analysis in Electrical Systems
12	Multiple Integration: 3D Modeling in Computer Graphics and AI
13	Fourier Series: Periodic Signal Analysis in Digital Systems
14	Half-range Expansions: Symmetry and Signal Compression Techniques
15	Harmonic Analysis: Frequency Components in Audio and Vibration Systems

BTECCE25102: Fundamentals of Electronics

Course Type	ESC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hrs./Week	Lecture	3	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	2Hrs./Week	Practical/Studio	1	Total Marks	100
Total	5 Hrs./Week	Total	4		

Course Description

This course will serve as an introduction to develop a foundational understanding of basic electronic components, analogue and linear circuits, instrumentation elements, and communication systems, enabling students to analyse, design, and evaluate simple electronic applications.

Course Outcomes

CO No.	At the end of this course students will be able to
1	Identify basic electronic components and devices.
2	Illustrate, explain and demonstrate working of simple electronic circuits based on Diodes, BJTs, FETs
3	Illustrate, explain and demonstrate working of simple Linear Integrated circuits using OP-AMPs.
4	Illustrate, explain and demonstrate working of elements in instrumentation system.
5	Distinguish between different communication systems and evaluate their characteristic requirements.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2	1					2					2	1			1	3
CO2	2	1					2					2				1	3
CO3	2	1					2					2				1	3
CO4	1	1					1					2	1			1	3
CO5	1	1					1					2				1	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	Introduction to Electronics Engineering and Circuits	9 Hours	CO	BTL
Introduction to different areas of Electronics and their applications in Engineering,, Diode applications- Half-wave rectifier, Full-wave rectifiers (Dual complementary bridge rectifier), Capacitor Filters, Zener diode-Voltage regulator.			1,2	3
Unit No.02	Bipolar Junction Transistor (BJT) Circuits	9 Hours	1,2	3
BJT types, structure, operation and Characteristics, Transistor in Common-Emitter CE, Transistor as a CE amplifier, Transistor as a switch.				
Unit No.03	Linear Integrated Circuits	9 Hours	3	3
Introduction to operational amplifiers, Block diagram of OP-AMP, Ideal parameters of OP-AMP, Negative feedback, Inverting & Non inverting Amplifier, Comparators, Summing amplifier, Difference amplifier.				
Unit No.04	Sensors and Transducers	9 Hours	4	3
Introduction to instrumentation system, Analog and digital data acquisition, Characteristics of Sensor and transducer, Types of sensor and transducer-Active & Passive Sensor, unipolar and bipolar sensor and transducers. RTD , Thermocouple, Strain gauge, LVDT.				
Unit No.05	Communication system	9 Hours	5	3
Importance of communication system, elements of communication system, bandwidth requirement, IEEE frequency spectrum, Transmission media: Wired (twisted cable, coaxial cable and optical fiber cables) and wireless, Need for modulation, analog modulation schemes: - AM, FM.				

Textbooks

Text Books
1. Floyd, "Electronic Devices and Circuits", Pearson Education.
2. Robert L. Boylestad, Louis Nashelsky-"Electronic Devices and Circuit Theory" -Prentice Hall
3. R P Jain," Modern Digital Electronics", TMH publications
4. H.S. Kalasi "Electronic Instrumentation", Tata McGraw Hill
5. Frenzel. "Communication Electronics. Principles and Applications." Third Edition. McGraw Hill Education.

Reference Books / Journal Articles / Weblink

Reference Books
6. Albert Malvino, "Electronic principles", 7thEd, TataMc-Graw-Hill.
7. John G. Webster, "The Measurement, Instrumentation & sensors Handbook", CRC Press.
8. Ramakant Gaikwad, "Linear Integrated Circuits", Pearson Education.

9. Anand Kumar, "Fundamental of digital circuits", prentice Hall of India.
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List of Experiments/Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Introduction of Electronic Components, Test and measurement instruments.	1	2
2	Observe the given AC voltage and its rectified output , output with capacitor filter of HWR and FWR	1,2	3
3	Zener diode characteristics , Zener as a voltage regulator	1,2	3
4	Single Stage BJT Common Emitter amplifier circuit. For a given BJT CE Amplifier circuit, a. Identify pins of a BJT (such as BC547) and study of its data sheet specifications. b. To measure voltages and observe waveforms at input and output terminals of Single stage BJT Common Emitter amplifier circuit. Calculate voltage gain of the amplifier	2	3
5	BJT as a switch	2	3
6	Op-amp based amplifiers circuits. Identify pins of an Op-amp (such as LM741). Implement Inverting & Non inverting DC Amplifiers.	3	3
7	Op-amp based amplifiers circuits-Summing amplifier and difference amplifier.	3	3
8	Build and Test an Instrumentation system with any one of the following sensors/ Transducers Resistive, Inductive Capacitive	4	3

Infosys Springboard Courses mapped.

Course Name	Contents	Links	CO
Electronics Course	Course Overview: This immersive Electronics course takes participants on a stimulating journey through the essential principles, components, and applications of electronics. Whether you're a novice with a curiosity for circuits or an aspiring engineer seeking to expand your skill set, this course caters to all levels of experience, fostering a strong foundation in electronics.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01373779399397376019/overview	1
Sensors and Sensor Circuit Design	After taking this course, you will be able to: Understand how to specify the proper thermal, flow, or rotary sensor for taking real-time process data, Implement thermal sensors into an embedded system in both hardware and software, Add the sensor and sensor interface into a microprocessor based development kit, Create hardware and firmware to process sensor signals and feed data to a microprocessor for further evaluation, Study sensor signal noise and apply proper hardware techniques to reduce it to acceptable levels..	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0142227350118318081281/overview	4
BJT	A bipolar junction transistor is a type of transistor that relies on the contact of two types of semiconductor for its operation. BJTs can be used as amplifiers, switches, or in oscillators. With this course, you will clear all the concepts of bipolar transistors(BJT)	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138417282215936002370_shared/overview	2

Semiconductor Electronics:Materials , Devices And Simple Circuits	Semiconductor Electronics:Materials, Devices And Simple Circuits	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01396008994203238454/overview	1
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BTECCE25103: Computer Programming Principles

Course Type	ESC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	5 Hr./Week	Total	4		

Course Description

Its introductory programming course for understanding of programming concepts and methods using C and C++ programming language.

Course Outcomes

CO No.	Statement
1	Apply programming fundamentals to construct concise solutions.
2	Demonstrate available algorithmic principles to generate efficient solutions.
3	Justify modular programming approach by making use of elementary as well as superior data structures & Implement programming constructs and derived data types to solve real-world problems.
4	Demonstrate the basic object-oriented programming concepts.
5	Apply advanced OOPs concepts in real world applications.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1		3	1										1		2		1,2
CO2	3	2		1	3									1	2		3
CO3	1	3							1					1		1	3,5
CO4	1	2		3									1	3	2	1	3,4
CO5			3	1	2				1							1	3,6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

B.Tech Computer Engineering 2025

Course Content

Unit No 1	Fundamentals of Computers & C Programming	Hours 9	CO	BTL
Computer Systems & Computing Environments, Computer Languages, Creating & Running Programs, Algorithms & Flowcharts, Introduction to C Language: History, Compiler, Linker, Library & Header Files, Structure of a C Program, Identifiers, Variables, Constants, Data Types, Basic Input/Output, Operators: Arithmetic, Relational, Logical, Bitwise, Expressions, Precedence & Type Conversions, Control Structures: if, switch, goto			1	1,2
Unit No 2	Loops, Arrays & Strings in C	Hours 9		
Loops: while, for, do-while, break, continue, Arrays: Declaration, Initialization, Memory Layout, 1D & 2D Arrays, Matrix Operations, Sparse Matrices, Character Arrays (Strings) & String Operations, Menu Driven Programs using Arrays & Strings			1,2	1,2,3
Unit No 3	Functions, Pointers & Structures in C	Hours 9		
Functions: Anatomy, Local & Global Scope, Nested Calls, Returning Values, Recursive Functions & Types, Pointer Concepts: Definition, Address, Dereferencing, Pointer Arithmetic, Pointers with Functions & Arrays, Structures: Definition, Declaration, Initialization, Array of Structures, Structure within Structure			1,2,3	1,2,4
Unit No 4	Introduction to C++ & OOP Concepts	Hours 9		
Procedure Oriented vs Object-Oriented Programming, Overview of OOP Principles (Abstraction, Encapsulation, Inheritance, Polymorphism), Introduction to C++, Program Structure, namespace, I/O (cin, cout), Identifiers, Variables, Constants, Operators, Typecasting, Control Structures in C++, Classes & Objects: Private/Public Members, Static Members, Constructors & Destructors			1,2,4	1,2,5
Unit No 5	Advanced OOP Features in C++	Hours 9		
Inheritance: Concept & Types (Single, Multiple, Multilevel, Hierarchical), Function Overloading, Inline Functions vs Macros, Friend Functions, Virtual Functions (Run-time Polymorphism), Data Abstraction & Encapsulation: Concepts, Benefits, Examples Exception Handling (try, catch, throw), Templates: Function & Class Templates			1,4,5	1,2,6

Textbooks

1	“Computer Fundamentals”, P. K. Sinha, Sixth Edition, BPB Publication, ISBN 81-7656-752-3.
2	“Let us C”, Y. Kanetkar, Second Edition, BPB Publication. ISBN: 8176566217.
3	E. Balagurusamy; “Object oriented programming with C++”; 4th Edition, Tata McGraw-Hill.

Reference Books / Journal Articles / Weblink

1	“Programming language – ANSI C”, Brain W Kernighan and Dennis Ritchie, Second edition ISBN 0-13-110370-9
2	“C By Example”, Kalicharan Noel, UK, Cambridge University Press.
3	“Programming with C- Schaum’s outline Series”, B. Gottfried, Second edition, Tata McGraw Hill Publication, ISBN 0-07-463491-7
4	“C and Data Structures”, E. Balaguruswamy, TMH Publication, 2003
5	R. Lafore, “The Waite Group's Object oriented Programming in C++”; 3rd Edition, Galgotia Publications.
6	E. Balagurusamy; “Programmng with Java”; 5th Edition, Tata McGraw-Hill.
7	Bjarne Stroustrup ;“ Object-Oriented Programming in C++ ” ;4th Edition, Sams Publishing.

List of Experiments/ Studio Assignments, etc.

1	Program to numbers such as finding smallest, largest number, conversion of degrees to Fahrenheit, etc.
2	Program on numbers such as finding GCD, LCM, Prime numbers, etc.
3	Program on displaying odd, Even nos. from a series or a range, Finding prime numbers in a series, etc.
4	Program for operations on 2D array such as accept, display, Add, multiply, transpose, etc.
5	Program on user defined string functions.
6	Program on recursive and non-recursive functions such as Factorial, Fibonacci series, etc.
7	Program on array of pointers and pointer to arrays.
8	Program on structures, unions and files to perform different operations
9	Write a C++ Program to display Names, Roll No., and grades of 3 students who have appeared in the examination. Declare the class of name, Roll No. and grade. Create an array of class objects. Read and display the contents of the array.
10	Write a C++ program to create Single level and multilevel inheritance.
11	Write a C++ program for Inline and virtual function.
12	Write a C++ program to illustrate the concept of Abstraction.
13	Write a program in C++ to implement the exception handling with multiple catch statements.
14	Write a C++ program to illustrate the concept of Template Class.

Springboard courses:

Unit	Topics	Springboard Course	CO
1. Fundamentals of Computers & C Programming	Computer Systems & Computing Environments, Computer Languages, Creating & Running Programs, Algorithms & Flowcharts, Introduction to C Language: History, Compiler, Linker, Library & Header Files, Structure of a C Program, Identifiers, Variables, Constants, Data Types, Basic Input/Output, Operators: Arithmetic, Relational, Logical, Bitwise, Expressions, Precedence & Type Conversions, Control Structures: if, switch, goto	<u>C Programming 101</u> <u>C Programming Course</u> <u>Programming in C</u> <u>Fundamentals of the C Language: Variables & Datatypes</u>	<u>1,2</u>
2. Loops, Arrays & Strings in C	Loops: while, for, do-while, break, continue, Arrays: Declaration, Initialization, Memory Layout, 1D & 2D Arrays, Matrix Operations, Sparse Matrices, Character Arrays (Strings) & String Operations, Menu Driven Programs using Arrays & Strings	<u>C Programming 101</u> <u>C Programming Course</u> <u>Handling Arrays in C</u> <u>Programming Two-Dimensional Arrays and Pointers in C</u> <u>Programming in C</u>	<u>1,2,3</u>
3. Functions, Pointers & Structures in C	Functions: Anatomy, Local & Global Scope, Nested Calls, Returning Values, Recursive Functions & Types, Pointer Concepts: Definition, Address, Dereferencing, Pointer Arithmetic, Pointers with Functions & Arrays, Structures: Definition, Declaration, Initialization, Array of Structures, Structure within Structure	<u>C Programming Course</u> <u>Working with Functions in C</u> <u>Program</u> <u>Programming in C</u> <u>Pointers In C</u> <u>Programming</u> <u>Two-Dimensional Arrays and Pointers in C</u>	<u>2,3</u>

4. Introduction to C++ & OOP Concepts	Procedure Oriented vs Object-Oriented Programming, Overview of OOP Principles (Abstraction, Encapsulation, Inheritance, Polymorphism), Introduction to C++, Program Structure, namespace, I/O (cin, cout), Identifiers, Variables, Constants, Operators, Typecasting, Control Structures in C++, Classes & Objects: Private/Public Members, Static Members, Constructors & Destructors	<u>C++ Fundamentals</u>	<u>4, 5</u>
5. Advanced OOP Features in C++	Inheritance: Concept & Types (Single, Multiple, Multilevel, Hierarchical), Function Overloading, Inline Functions vs Macros, Friend Functions, Virtual Functions (Run-time Polymorphism), Data Abstraction & Encapsulation: Concepts, Benefits, Examples Exception Handling (try, catch, throw), Templates: Function & Class Templates	<u>Programming using C++</u>	<u>4, 5</u>

Projects on Infosys Springboard:

1. Project in C Programming - Hangman Game
2. Project on C Programming - Payroll Processing
3. Project in C Programming - Employee Management System
4. Project in C Programming - 2 Players Tic-Tac-Toe Game

BTECAD25104/BTECAI25104: Web Application Development

Course Type	VSEC	Semester	1
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Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hr./Week	Lecture	1	CIE Marks	
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	3 Hr./Week	Total	2		

Course Description

This course helps the student to understand the principles and methodologies of web-based application development process and current client-side scripting technologies

Course Outcomes

CO No.	At the end of this course students will be able to
1	Analyze given assignment to select sustainable web development, design methodology
2	Develop web-based application using suitable client-side web technologies
3	Enable to design and develop solutions to problems using technologies studied.
4	Design and develop solution to handle information which is location independent.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3																1,2
CO2				3													1,2,3
CO3							3										1,2,3,6
CO4													3				1,2,3,6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction & HTML	Hours 4
Evolution of Web, Web Browser, Web Server, Application server, Client Server Architecture and Communication, Client Server communication techniques, URL, URI, web design concepts and issues, HTTP request response structure of html document, elements: headings, paragraphs, line break, colors& fonts, links, frames, lists, tables, images and forms, page division using div flex		
Unit No 2	Cascading Style Sheets	Hours 3
DTD: Schema, elements, attributes, Benefits, syntax, inline, internal, external style sheets, CSS selectors (id, class, grouping, child), display behavior and positioning (static, fixed, relative, absolute, floats), visibility of element attribute, adding scrollbars, Modern CSS (Flexbox, Grid)		
Unit No 3	JavaScript & Server-Side Scripting	Hours 5
HTML (Embedded, External), variable declarations, operators, Control statements, Arrays, Functions and Scopes, Objects in Java Scripts, regular expressions for input validations, error handling DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM Introduction to Server-Side Programming, Core Concepts: HTTP, Request/Response, RESTful APIs, Server		
Unit No 4	E-commerce Web Application Development	Hours 3
Introduction to E-commerce (Types, Models: B2C, B2B, C2C), Essential E-commerce Features (Catalog, Cart, Checkout, Payment Integration), UI/UX Principles for Web Applications		

Textbook

1	<i>Achyut Godbole & Atul Karate, Web Technologies: TCP/IP to Internet Application Architectures</i> , McGraw Hill Education publications, ISBN, 007047298X, 9780070472983
2	<i>Ralph Moseley & M. T. Sav Aliya, —Developing Web Applications</i> , Wiley publications, ISBN 13: 9788126538676

Mini Project

- Students have to implement mini project which should include following concepts: HTML, CSS, JavaScript

List of Practice Assignments:

- Design web pages which will include concepts of HTML
- Apply CSS to the created web pages.
- Write java script where ever required for web pages created or for input validation

Suggestions:

- Read and Display syllabus structure, add filter option
- Design page to get student details information with validation
- Design pages to display food information with images

Unit	Topics	Springboard Course	CO
Introduction & HTML	Evolution of Web, Web Browser, Web Server, Application server, Client Server Architecture and Communication, Client Server communication techniques, URL, URI, web design concepts and issues, HTTP request response structure of html document, elements: headings, paragraphs, line break, colors& fonts, links, frames, lists, tables, images and forms, page division using div flex	What's the World Wide Web (WWW) And How it Works https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0133013212890316807911_shared/overview Introduction to HTML https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138417493460582402881_shared/overview HTML Beginners https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01384323834364723233959_shared/overview HTML Tutorial https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0142354068337541123166/overview	1
Cascading Style Sheets	DTD: Schema, elements, attributes, Benefits, syntax, inline, internal, external style sheets, CSS selectors (id, class, grouping, child), display behavior and positioning (static, fixed, relative, absolute, floats), visibility of element attribute, adding scrollbars, Modern CSS (Flexbox, Grid)	Learning CSS https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0130944328695070722286_shared/overview Practical HTML and CSS https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01330386999904665628599_shared/overview	2
JavaScript & Server-Side Scripting	HTML (Embedded, External), variable declarations, operators, Control statements, Arrays, Functions and Scopes, Objects in Java Scripts, regular expressions for	Javascript https://infyspringboard.onwingspan.com/web/en/app/toc/lex_18109698366332810000_shared/overview	3

	input validations, error handling DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM Introduction to Server-Side Programming, Core Concepts: HTTP, Request/Response, RESTful APIs, Server	Javascript Tutorial https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014222738523209728949/overview	
E-commerce Web Application Development	Introduction to E-commerce (Types, Models: B2C, B2B, C2C), Essential E-commerce Features (Catalog, Cart, Checkout, Payment Integration), UI/UX Principles for Web Applications	Introduction to Web Application Development https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013842618018390016188_90shared/overview Introduction To UI/UX https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0135015589402624008773/overview	4

BTECCE25105: Applied Science and Engineering Project I

Course Type	BSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	04		

Course Description

This course is focusing on understanding the core principles and mindset that drive design thinking. This course will be used to complete the design thinking process to tackle real-world challenges and discover its business applications by creative and critical thinking.

Course Outcomes

CO No.	Statement
1	CO1: Identify and define real-world problems through stakeholder analysis, literature review, and benchmarking across societal and scientific domains.
2	CO2: Develop and implement solution strategies—products, systems, or experiments—using appropriate tools, resources, and interdisciplinary knowledge.
3	CO3: Communicate and refine project outcomes through iterative prototyping, documentation, peer feedback, and structured project planning (e.g., Gantt charts, risk analysis).

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	1	1	3			1	1	1	2	1	1	1					1,2,3
CO2						2	2	1	1	2	1	1					2,3
CO3							3			1		2					1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	Project Execution Guidelines	30 Hours	CO	BTL
1.	Introduction to design and scientific problem-solving		CO1	1,2,3
2.	Brainstorming issues across domains: environment, health, energy, mobility, etc.		- 1 to 7	
3.	Team/individual project selection and Stakeholder identification (e.g., users, community, clients)		CO2	
4.	Literature review and benchmarking		- 8 to 11	
5.	Literature review and benchmarking - continue			
6.	Literature review and benchmarking			
7.	Creative and collaborative problem solving – open ended discussion		CO3-	
8.	Finalizing the problem scope with constraints and impact and translate problem into clear objectives or testable hypotheses		9 to 15	
9.	Choose solution format: product, process, system, app, experiment and identify needed knowledge/skills (coding, circuits, design, math, etc.)			
10.	Learn selected tools (e.g., Python, Arduino, CAD, Excel, Power BI, ML tools)			
11.	Resource planning: materials, software, tools			
12.	Prepare timelines using Gantt charts and Risk analysis and contingency planning			
13.	Initial prototype building (physical/software) or setting up experiments			
14.	Iterative refinement: making small changes based on observations			
15.	Documentation of technical process and Peer feedback on prototype/experiment setup			

Textbooks

1	
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Reference Books / Journal Articles / Weblink

Reference Books	
1	https://www.linkedin.com/posts/kamal-shahid-0126_read-10-papers-in-1-hour-activity-7346830717152022528-LWeQ?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTIUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/ali-mk-hindi-%F0%9F%87%AE%F0%9F%87%B6%F0%9F%87%B5%F0%9F%87%B8-4b1b80115_all-the-research-tools-you-need-in-2025-activity-7341087359205007362-

8kdX?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTIUyeChOpLEy5VudBSDA https://www.linkedin.com/posts/drasadnaveed_need-help-choosing-a-new-research-topic-activity-7347110335511674880-MU8u?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTIUyeChOpLEy5VudBSDA
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BTECCCE25106: Value Added Career Orientation I

Course Type	CC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	02		

Course Description

To empower F.Y.B.Tech students with career readiness, soft skills, and holistic personal development through collaborative exploration of resources, group projects, and team-based creative assessments.

Course Outcomes:

Upon completion of this course, the student will be able to:

CO No.	Statement
1	Reflect on your strengths, interests, and learning style, and begin to plan your academic and career journey
2	Identify different types of careers, work models, and new trends like remote work, gig economy, and use of technology in jobs.
3	Understand how to work with people from different cultures and industries and learn basic teamwork and workplace skills.
4	Explore how new ideas are developed through research and innovation and understand the basics of entrepreneurship.
5	Recognize the importance of giving back to society, volunteering, and learning about careers in government and social sectors.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1								1	1			3		1		1	4
CO2					1	1					2				1	1	1
CO3							1	1	1		1			1			2
CO4						3	2			2			1		1		3
CO5							1	1		3	2				1	1	2, 4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	LEARNING STRATEGIES & ACADEMIC PROGRESSION	06 Hours	CO	BTL
Self-awareness- values, interests, personality, and purpose; Career-aligned academic planning (UG+PG, India & abroad); Learning how to learn- Growth mindset, metacognition, self-regulation; Building transferable skills- Communication, critical thinking, collaboration, digital fluency; Navigating interdisciplinary learning and lifelong learning paths			1	1, 2
UNIT NO.02	CAREER ORIENTATION & FUTURE OF WORK	06 Hours		
Defining a career; Career vs. job vs. livelihood; Evolving work models- Portfolio careers, gig economy, hybrid & remote work; Career success in the 21st century- Purpose, flexibility, and impact; Leveraging technology- AI, automation, and digital transformation; Interdisciplinary thinking and ESG awareness in shaping careers			2	2, 3
UNIT NO.03	GLOBAL CITIZENSHIP & INDUSTRY ENGAGEMENT	06 Hours		
Global citizenship and intercultural fluency; Working in global teams and across cultures; Teamwork, collaboration, conflict resolution, and inclusion; Introduction to business sectors and industries; Understanding corporate etiquette and workplace dynamics			3	3, 4
UNIT NO.04	INNOVATION & ENTREPRENEURSHIP	06 Hours		
Curiosity and problem-solving through research; Innovation ecosystems- Labs, incubators, hackathons; Entrepreneurial mindset and social enterprise; From ideas to action: Generating, refining, pitching; Learning from real-world examples- Startup or innovation journeys			4	2, 3
UNIT NO.05	COMMUNITY ENGAGEMENT & GOVERNMENT/PUBLIC SECTOR	06 Hours		
Civic responsibility and active citizenship; Volunteerism and service-learning; Understanding social innovation and grassroots impact; Engaging with local communities and marginalized voices; Public policy awareness and sustainable development goals (SDGs)			5	4, 5

Reference Books:

1.	“What Color Is Your Parachute? for College: Pave Your Path from Major to Meaningful Work” by Katherine Brooks, Ten Speed Press (2021).
2.	"The Start-up of You: Adapt, Take Risks, Grow Your Network" by Reid Hoffman and Ben Casnocha, Random House Business (2013).
3.	“Creative Confidence: Unleashing the Creative Potential Within Us All” by Tom Kelley and David Kelley (2013).

Infosys Springboard Courses:

Unit	Topics	Springboard Course	CO
Learning Strategies & Academic Progression	Metacognition, self-regulation, study habits	Learning How to Learn https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0135015651011051529522/overview	1
Career Orientation & Future of Work	Industry trends, automation, tech impact	Future of Work https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01435729566264524837/overview	2
Global Citizenship & Industry Engagement	Collaboration, cross-cultural communication, inclusion	Interpersonal Skills https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013896956528287744589_shared/overview	3
Innovation & Entrepreneurship	Small business planning and strategy	Entrepreneurship-Introduction to Business Model Canvas https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01384347494667878439748_shared/overview	4
Community Engagement & Government/Public Sector	Innovation	Innovation https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013782185588604928167/overview	5

IKS25CE01A: Indian Knowledge System (IKS)

Course Type	IKS	Semester	1
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hrs./Week	Lecture	2	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	2 Hrs./Week	Total	2		

Course Discription

The purpose of the course is to provide a general introduction to Indian Knowledge System (IKS) and sensitize the students to the contributions made by ancient Indians in the field of Science, Philosophy and related applications and concepts.

Course Outcomes

CO No.	At the end of this course students will be able to
1	Understand the basic of IKS and Vedic corpus.
2	Understand Mathematical Sciences without knowing sciences whereas, engineering.
3	Understand Astronomy and Engineering technology in metals and metal working
4	Understand the need for an interface. Compare and contrast memory mapping and IO mapping techniques. Understand the Applications in Engineering and Technology
5	Understand Language , Knowledge Framework and linguistic

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	1					2						3	1				,3
CO2		2				1						3	1				2,3
CO3			1			2						3	1				2
CO4			1			2						3	1				2
CO5			1			2						3	1				1,2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	Indian Knowledge System – An Introduction and Vedic Corpus	6 Hours	CO	BTL
IKS Introductio,Organization of IKS,Historicity of IKS,Some salient aspects of IKS, Introduction to Vedas,A synopsis of the four Vedas,Sub-classification of Vedas,Messages in Vedas,Introduction to Vedāṅgas Prologue on Śikṣā and Vyākaraṇa,Basics of Nirukta and Chandas,Introduction to Kalpa and Jyotiṣa,Vedic Life: A Distinctive Features			1	2
Unit No.02	Number Systems and Units of Measurement and Mathematics	6 Hours	2	2.3
Number Systems and Units of Measurement: Number systems in India - Historical evidence,Salient aspects of Indian Mathematics,Bhūta-Saṃkhyā system,Kaṭapayādi system,Measurements for time, distance, and weight Piṅgala and the Binary system Mathematics: Introduction to Indian Mathematics,Unique aspects of Indian Mathematics Indian Mathematicians and their Contributions Algebra Geometry Trigonometry Binary mathematics and combinatorial problems in Chandaḥ Śāstra Magic squares in India				
Unit No.03	Astronomy and Engineering and Technology: Metals and Metalworking:	6 Hours	3	2
Astronomy: Introduction to Indian astronomy,Indian contributions in astronomy,The celestial coordinate system.,Elements of the Indian calendar Notion of years and months,Pañcāṅga – The Indian calendar system, Astronomical Instruments (Yantras),Jantar Mantar of Rājā Jai Singh Sawai Engineering and Technology: Metals and Metalworking: Wootz Steel: The rise and fall of a great Indian technology,The Indian S & T heritage ,Mining and ore extraction,Metals and metalworking technology , Iron and steel in India ,Lost wax casting of idols and artefacts,Aparatuses used for extraction of metallic components				
Unit No.04	Engineering and Technology: Other applications and Town Planning and Architecture:	6 Hours	4	2
Engineering and Technology: Other applications: Irrigation systems and practices in South India , Literary sources for science and technology ,Physical structures in India ,Irrigation and water management, Dyes and painting technology, The art of making perfumes Surgical techniques , Shipbuilding , Sixty-four art forms (64 Kalās) , Status of Indigenous S & T. Town Planning and Architecture: Perspective of Arthaśāstra on town planning ,Vāstu-śāstra – The science of architecture ,				

Eight limbs of Vāstu ,Town planning ,Temples in India: marvelous stone architecture for eternity , Temple architecture in India Iconography				
Unit No.05	Knowledge Framework and classifications and Linguistics	6 Hours	5	1,2
Knowledge Framework and classifications: Indian scheme of knowledge , The knowledge triangle , Prameya – A vaiśeṣikan approach to physical reality , Dravyas – the constituents of the physical reality , Attributes – the properties of substances and Action – the driver of conjunction and disjunction , Sāmānya, viśeṣa, samavāya Pramāṇa – the means of valid knowledge ,Saṁśaya – ambiguities in existing knowledge , Framework for establishing valid knowledge Deductive or inductive logic framework, Potential fallacies in the reasoning process ,Siddhānta: established tenets in a field of study Linguistics Introduction to Linguistics , Aṣṭādhyāyī ,Phonetics , Word generation Computational aspects , Mnemonics , Recursive operations , Rule based operations , Sentence formation , Verbs and prefixes , Role of Sanskrit in natural language processing				

Textbooks:

1	Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavana R.N. (2022), “Introduction to Indian Knowledge System: Concepts and Applications”, PHI Learning Private Ltd. Delhi.
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Reference Books / Journal Articles / Weblink

Reference Books	
1	Pride of India: A Glimpse into India’s Scientific Heritage, Samskrita Bharati, New Delhi.
2	Sampad and Vijay (2011). “The Wonder that is Sanskrit”, Sri Aurobindo Society, Puducherry.
3	Bag, A.K. (1979). Mathematics in Ancient and Medieval India, Chaukhamba Orientalia, New Delhi.
4	Datta, B. and Singh, A.N. (1962). History of Hindu Mathematics: Parts I and II, Asia Publishing House, Mumbai.
5	Kapoor Kapil, Singh Avadhesh (2021). “Indian Knowledge Systems Vol – I & II”, Indian Institute of Advanced Study, Shimla, H.P.
6	Subbarayappa, B.V. and Sarma, K.V. (1985). Indian Astronomy: A Source Book, Nehru Centre, Mumbai